

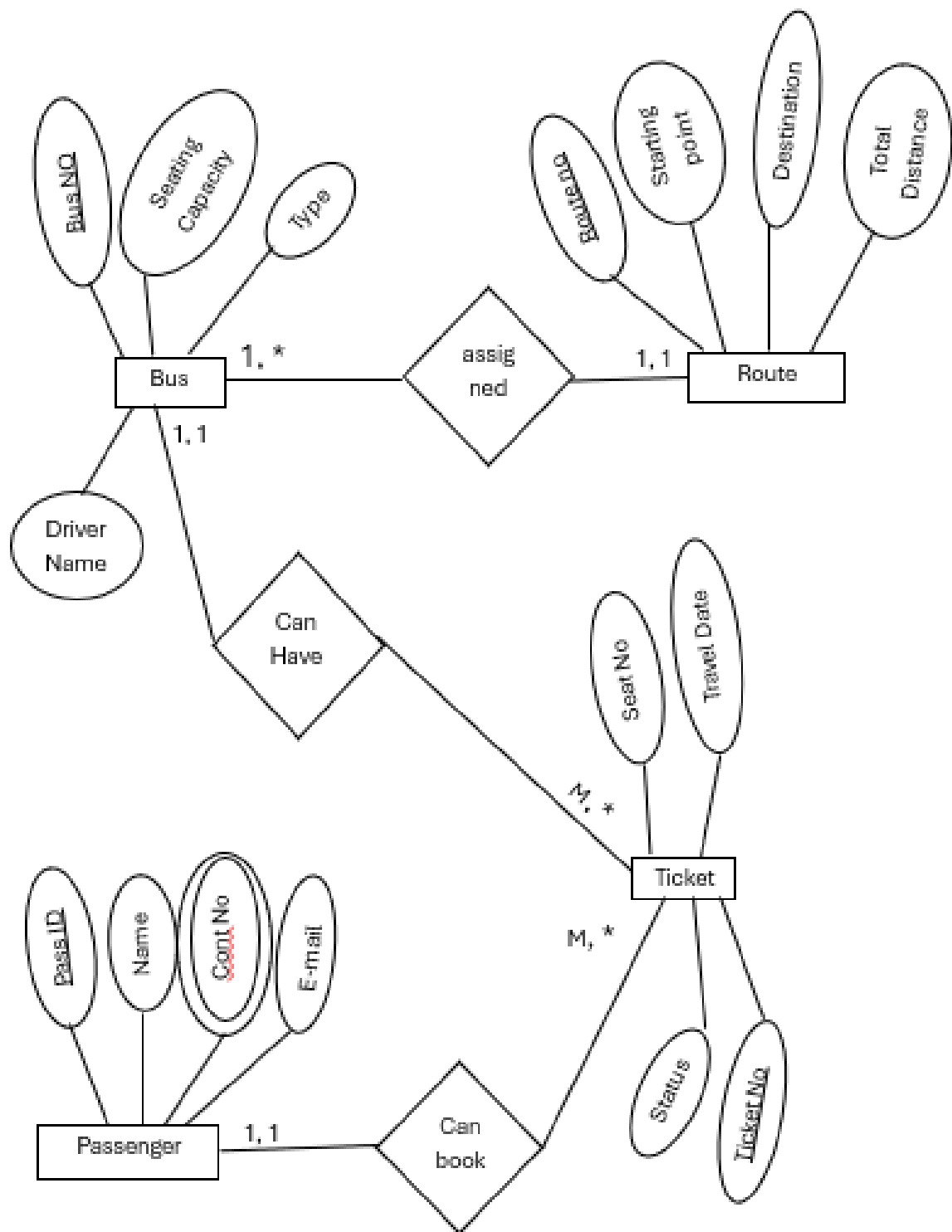
Case Study / Scenario

Student ID1: 23-52361-2 Name: NAYEM SIDDIKI	StudentID3: 23-50713-1 Name: MD. ABDUR RAUF
StudentID2: 23-52314-2 Name: FARHAN CHOWDHURY	StudentID4: 23-52586-2 Name: MD AL MURSALIN
CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

Bus Ticket Management System

In a Bus Ticket Management System, buses are identified by details such as a Bus number, seating capacity, type (like AC or non-AC), and the driver's name. Each bus is assigned to a specific route, which includes information like route NO, starting point, destination and total distance. Passengers are identified by their Passenger ID, along with personal details such as their name, contact number and e-mail. These passengers can book tickets for different bus routes. Each ticket is associated with a particular passenger and contains important details such as a ticket number, the travel date, the seat number assigned, and the status of the ticket (whether it's booked, confirmed, or canceled). The system ensures that each bus operates on a specific route, while passengers can book multiple tickets for different journeys. Multiple passengers can book seats on the same bus, and the system organizes all the information, ensuring smooth travel arrangements.

ER Diagram



Normalization

1NF Table

Ticket_No	Bus_No	Seating_Capacity	Type	Driver_Name	Route_No	Start_Point	Destination	Passenger_ID	Passenger_Name	Contact_No	Travel_Date	Seat_No	Status
T001	001	50	AC	John Doe	101	City A	City B	P001	Alice Green	1234567890	2024-09-18	10	Confirmed
T002	001	50	AC	John Doe	101	City A	City B	P002	Bob Brown	0987654321	2024-09-18	12	Booked
T003	002	40	Non-AC	Jane Smith	102	City C	City D	P003	Carol White	1122334455	2024-09-19	5	Confirmed
T004	002	40	Non-AC	Jane Smith	102	City C	City D	P004	David Black	6677889900	2024-09-19	8	Canceled
T005	003	55	AC	Mike Green	103	City E	City F	P005	Eve Brown	2233445566	2024-09-20	15	Booked

2^{NF} Table

Table: 1

Ticket_No	Passenger_ID	Travel_Date	Seat_No	Status	Bus_No	Route_No	Passenger_ID	Passenger_Name	Contact_No
T001	P001	2024-09-18	10	Confirmed	001	101	P001	Alice Green	1234567890
T002	P002	2024-09-18	12	Booked	001	101	P002	Bob Brown	0987654321
T003	P003	2024-09-19	5	Confirmed	002	102	P003	Carol White	1122334455
T004	P004	2024-09-19	8	Canceled	002	102	P004	David Black	6677889900
T005	P005	2024-09-20	15	Booked	003	103	P005	Eve Brown	2233445566

Table: 2

Bus_No	Seating_Capacity	Type	Driver_Name	Route_No	Start_Point	Destination	Total_Distance
001	50	AC	John Doe	101	City A	City B	200 km
002	40	Non-AC	Jane Smith	102	City C	City D	150 km
003	55	AC	Mike Green	103	City E	City F	300 km

Finalization

3NF Table

Table 1: Ticket Table

Ticket_No	Passenger_ID	Travel_Date	Seat_No	Status	Bus_No	Route_No
T001	P001	2024-09-18	10	Confirmed	001	101
T002	P002	2024-09-18	12	Booked	001	101
T003	P003	2024-09-19	5	Confirmed	002	102
T004	P004	2024-09-19	8	Canceled	002	102
T005	P005	2024-09-20	15	Booked	003	103

Table 2: Bus Table

Bus_No	Seating_Capacity	Type	Driver_Name
001	50	AC	John Doe
002	40	Non-AC	Jane Smith
003	55	AC	Mike Green

Table 3: Route Table

Route_No	Start_Point	Destination	Total_Distance
101	City A	City B	200 km
102	City C	City D	150 km
103	City E	City F	300 km

Table 4: Passenger Table

Passenger_ID	Passenger_Name	Contact_No
P001	Alice Green	1234567890
P002	Bob Brown	0987654321
P003	Carol White	1122334455
P004	David Black	6677889900
P005	Eve Brown	2233445566